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Image Restoration via Optimal Control of Nonlinear Fractional Systems

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Abstract. This study investigates a novel nonlinear optimization framework for image denoising under mixed noise conditions, incorporating a partial differential equation (PDE) constraint. By combining an H^{-s} -norm regularization with a decomposition strategy, we formulate a nonlinear system involving the fractional Laplacian operator. The existence and uniqueness of weak solutions to the associated direct problem are established using Schauder's fixed-point theorem. Furthermore, the well-posedness of the resulting optimal control problem is analyzed, and the existence of weak solutions to the corresponding adjoint problem is proved via the Galerkin method.

To numerically approximate the solution of the proposed model, we introduce an appropriate discretization scheme together with a primal–dual algorithm. Finally, a series of comparative numerical experiments is conducted to assess the efficiency, robustness, and effectiveness of the proposed approach in mixed-noise image denoising.

Key words: constrained PDE, fractional-order PDE, image denoising, nonlinear diffusion .

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